Optimality of Public Persuasion for Single-Good Allocation

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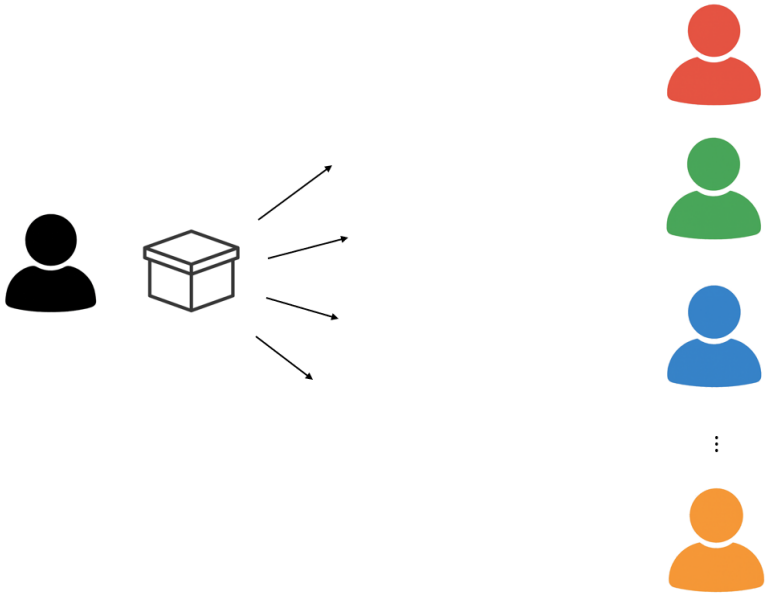
Model

- A sender allocates **an indivisible good** among n receivers
  - Sender can allocate to at most one receiver
  - Each receiver decides whether to accept based on self-interest



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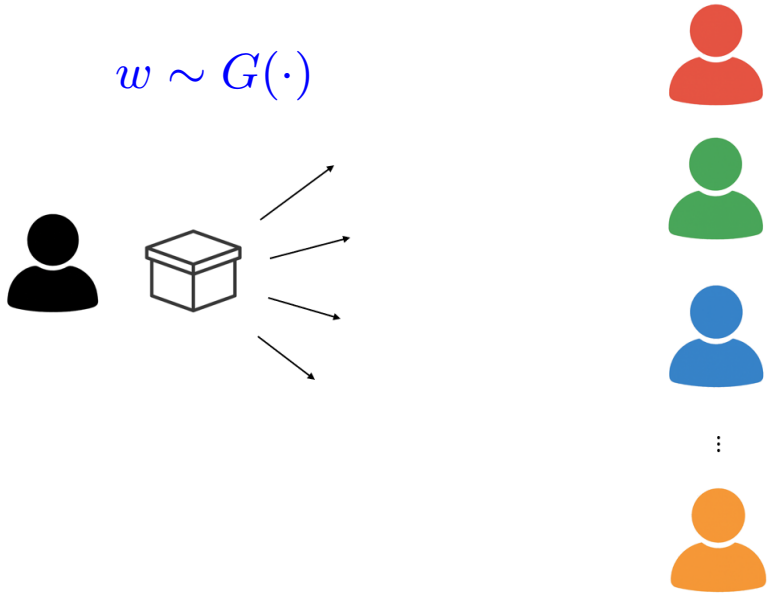

- Sender strategically discloses good's characteristics $w \in \Omega$, with prior dist $G(\cdot)$

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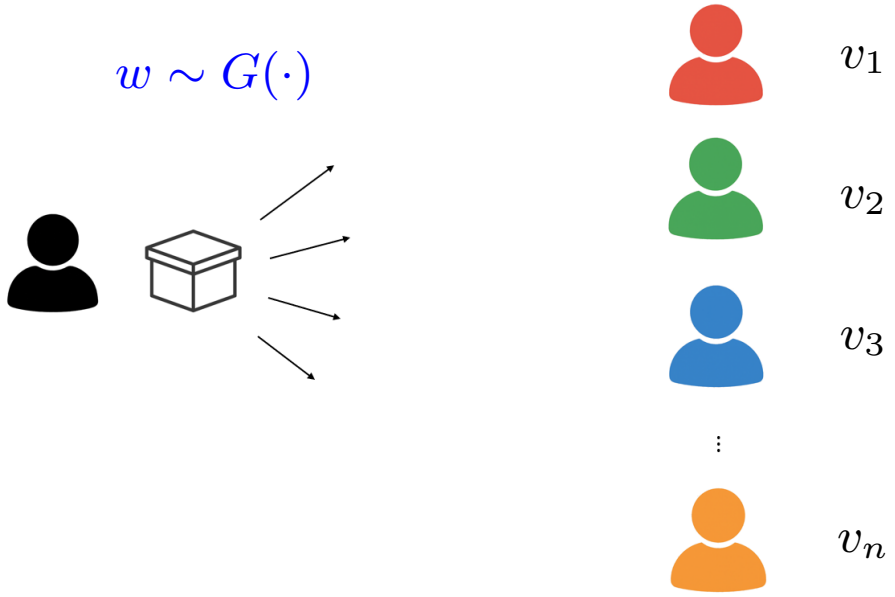

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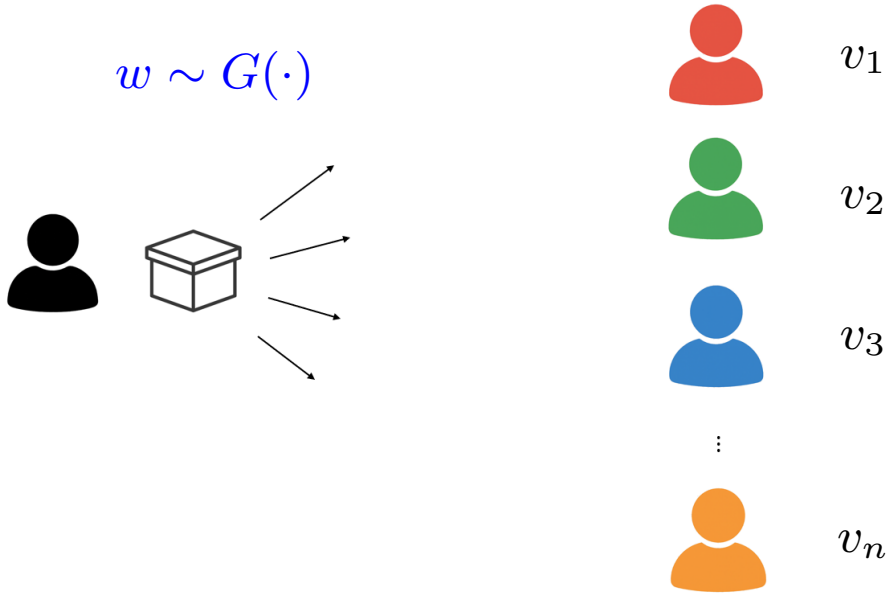

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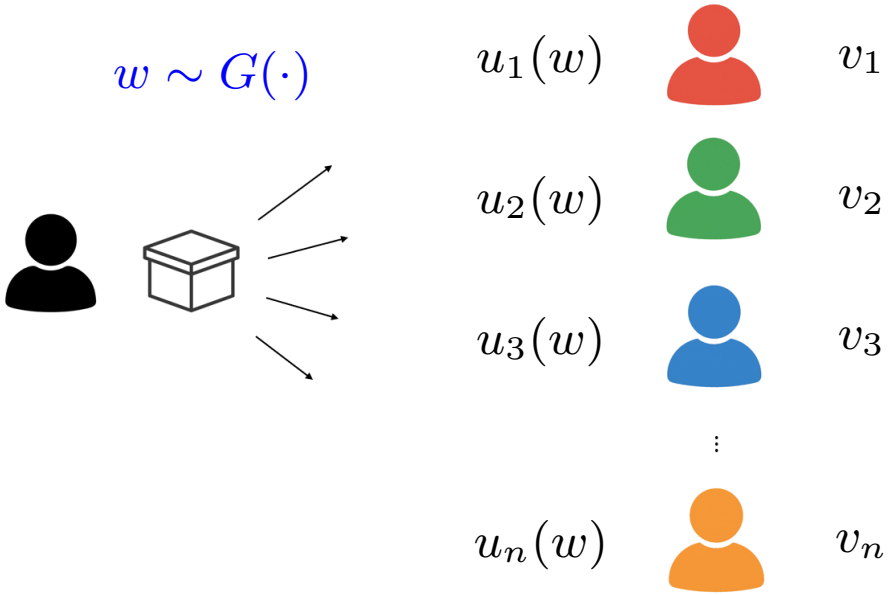

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- **Receiver i's utility:** $u_i(w)$ from receiving good of chars w; zero otherwise

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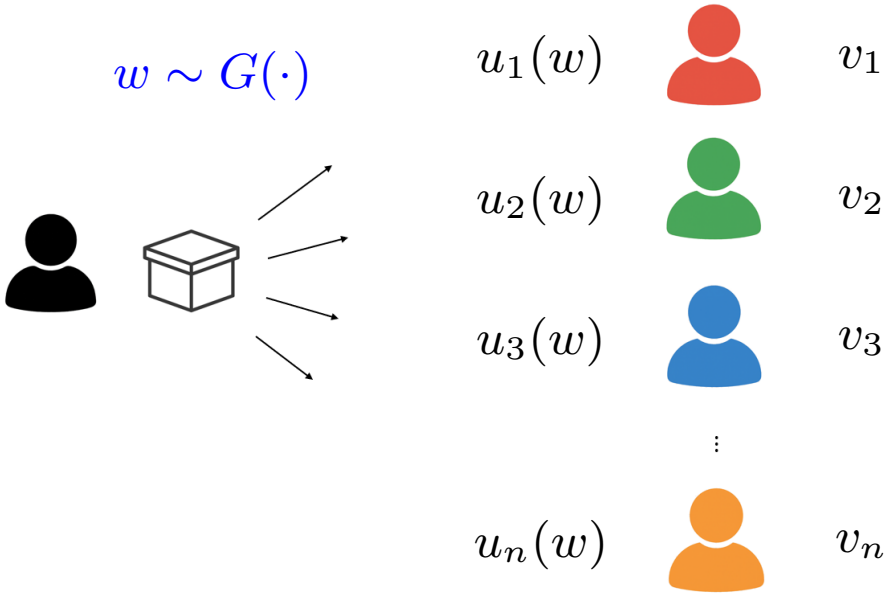

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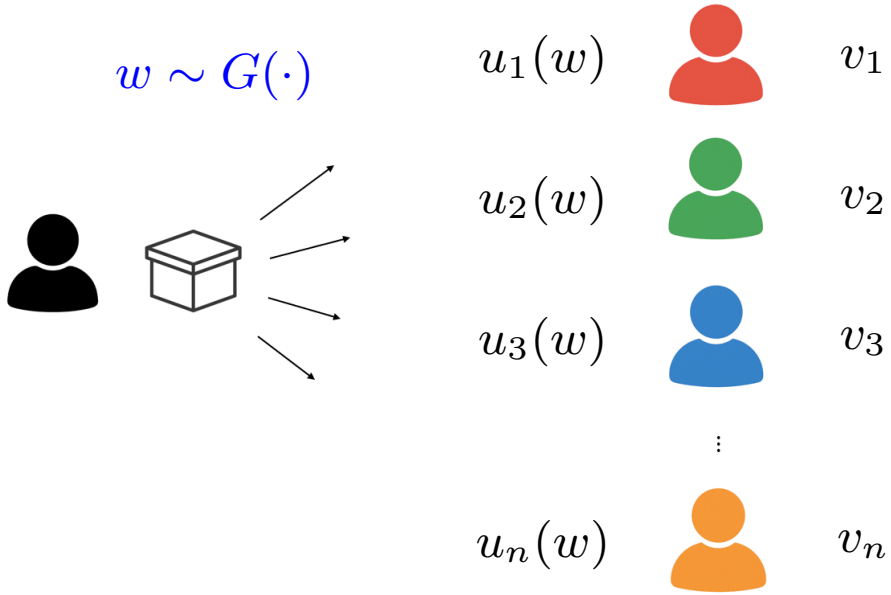
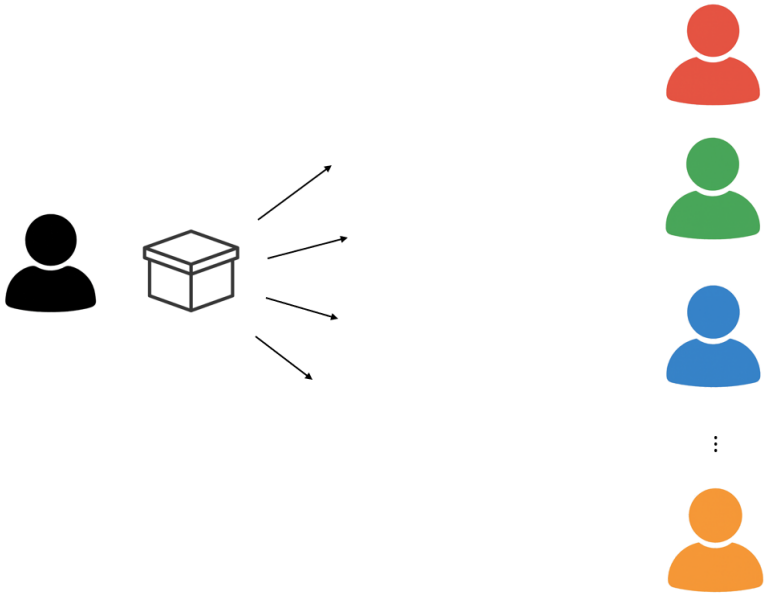


- **Sender's decision:** strategically reveal information about w (more below)
- **Receiver's decision:** whether to accept the good (or extend an offer)
- **Applications**: (i) school advisor promotes student for job positions, (ii) incubator pitches startup to VC investors, (iii) biotech seeks partnership with pharma companies.

Public versus private persuasion

Sender commits to persuasion mechanism $f(\boldsymbol{s}|w)$: joint dist of sending signals $\boldsymbol{s} = (s_1, s_2, \cdots, s_n)$ conditional on w



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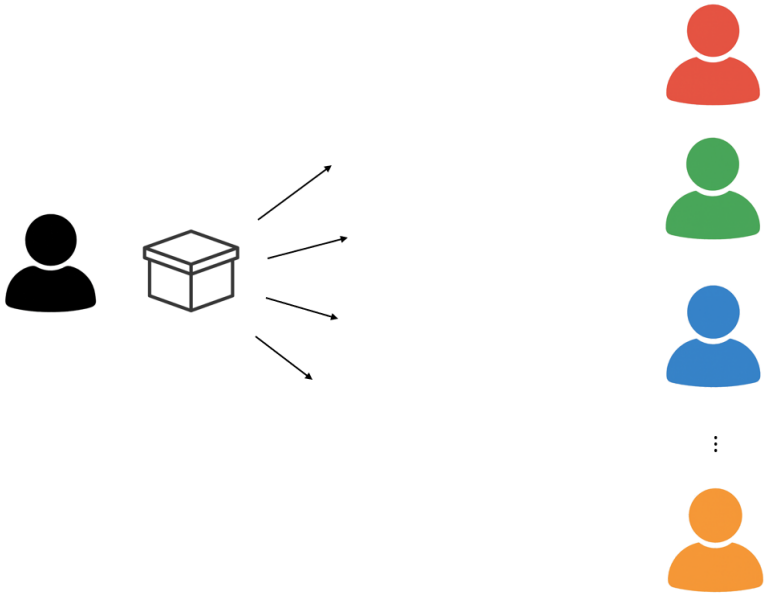

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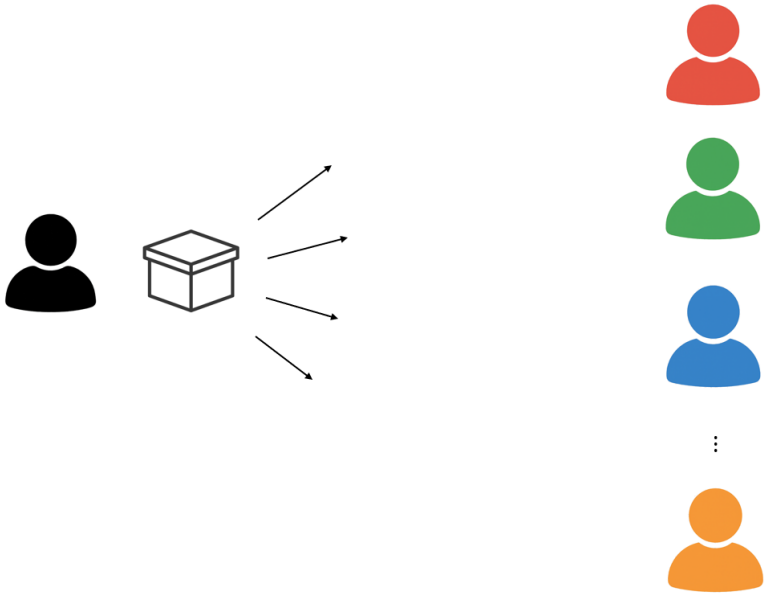

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- Model permits receivers to **communicate** after receiving signals
  - Receivers can communicate in an arbitrary way (including in self-interest)

Model (cont.)

The game proceeds as follows:

1. Sender commits to a persuasion mechanism $f(\cdot|w)$ and a signal space $\mathbf{S} = \bigotimes_{i=1}^{n} S_i$.
2. Sender observes the good's characteristics $w \sim G(w)$. A signal $\mathbf{s} = (s_i)_{i \in [n]} \sim f(\cdot|w)$ is generated and sent to the receivers.
3. Receivers communicate with one another in a certain way.
4. Each receiver i decides whether to accept the good based on the signal and communication.
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Main result: Public persuasion is optimal regardless of how receivers communicate

$\Longrightarrow$ Sender eliminates any communication for her own interest

Outline

- Motivation and problem formulation
- Optimality of public persuasion
- Extensions
- Structural properties
- Conclusions

The first-best relaxation

Proposition. The optimal value $\bar{V}$ of:

$$\max_{q(i|w)\geq 0} \quad \sum_{i=1}^{n} v_i \int_{w\in\Omega} q(i|w)\, dG(w)$$

$$\text{s.t.} \quad \int_{w\in\Omega} u_i(w)\, q(i|w)\, dG(w) \geq 0,\ \forall\, i \in [n], \quad \textbf{(participation constr.)}$$

$$\sum_{i\in[n]} q(i|w) \leq 1,\ \forall\, w \in \Omega.$$

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Proof sketch: let $q(i|w)$ be the allocation probabilities under equilibrium.

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$\Longrightarrow$ Public persuasion is optimal

Suboptimality of vanilla private persuasion

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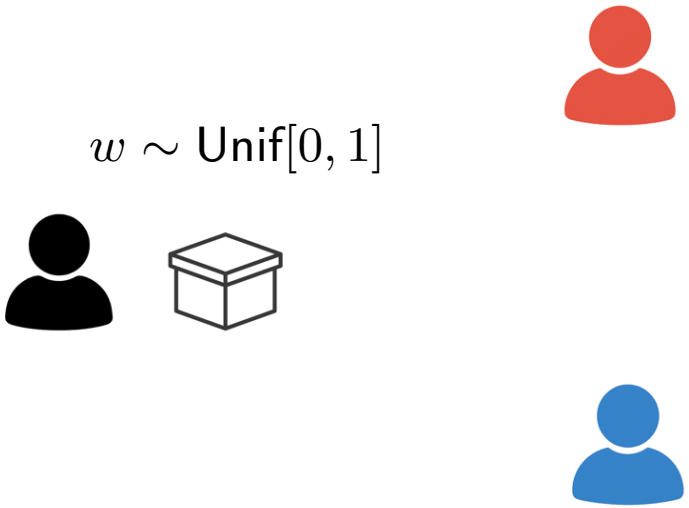

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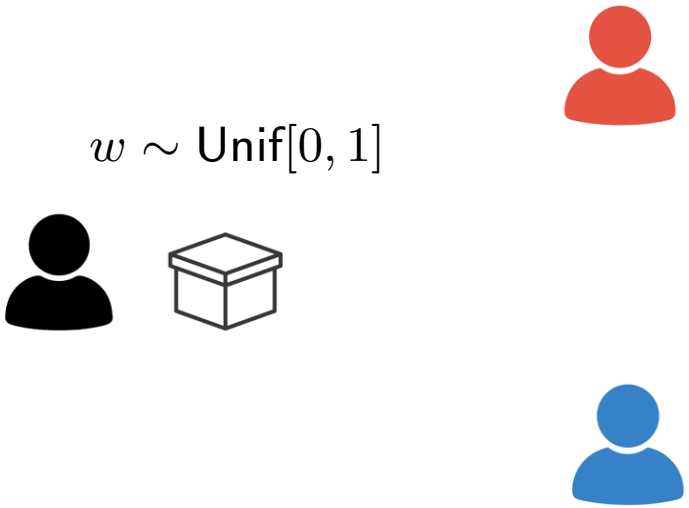

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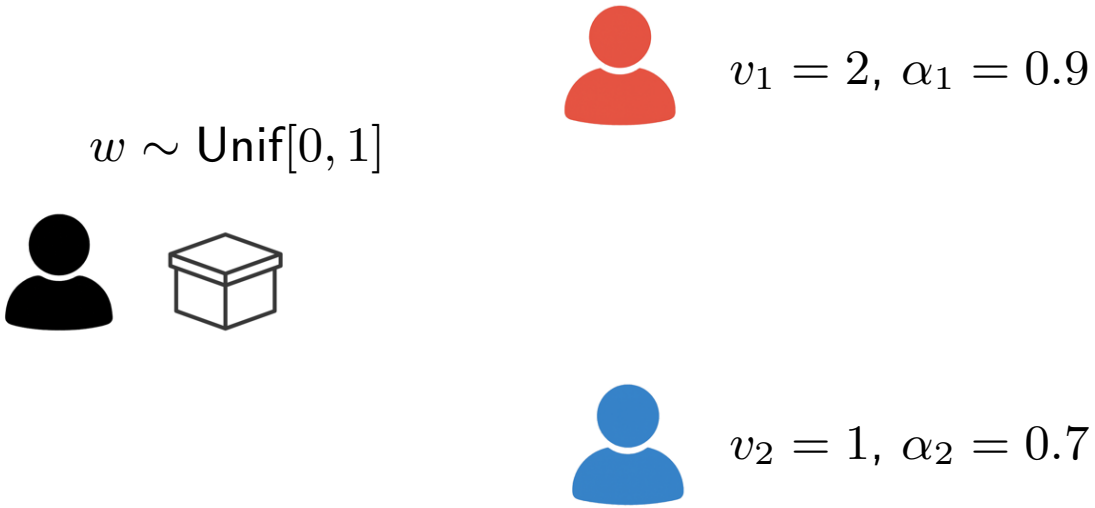

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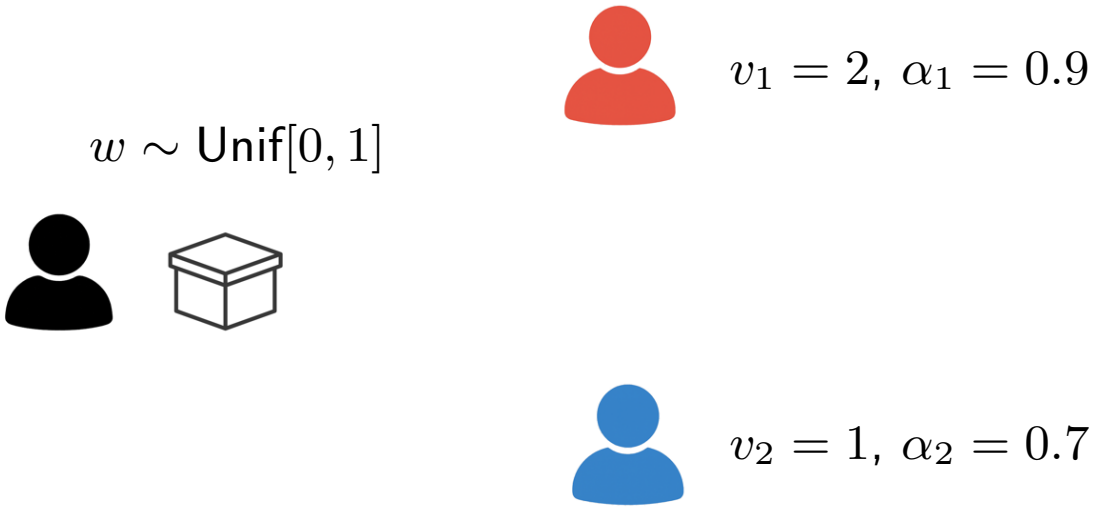

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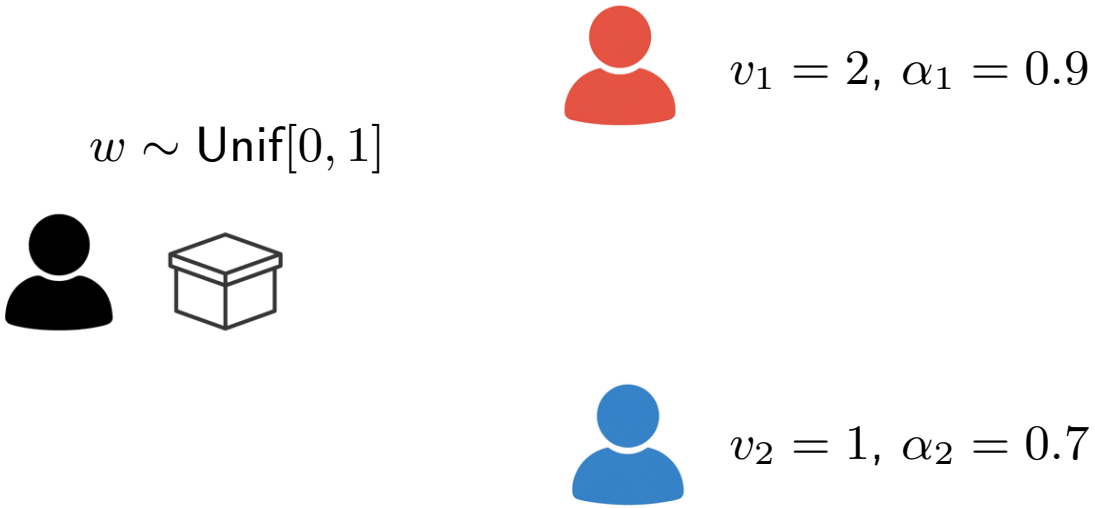

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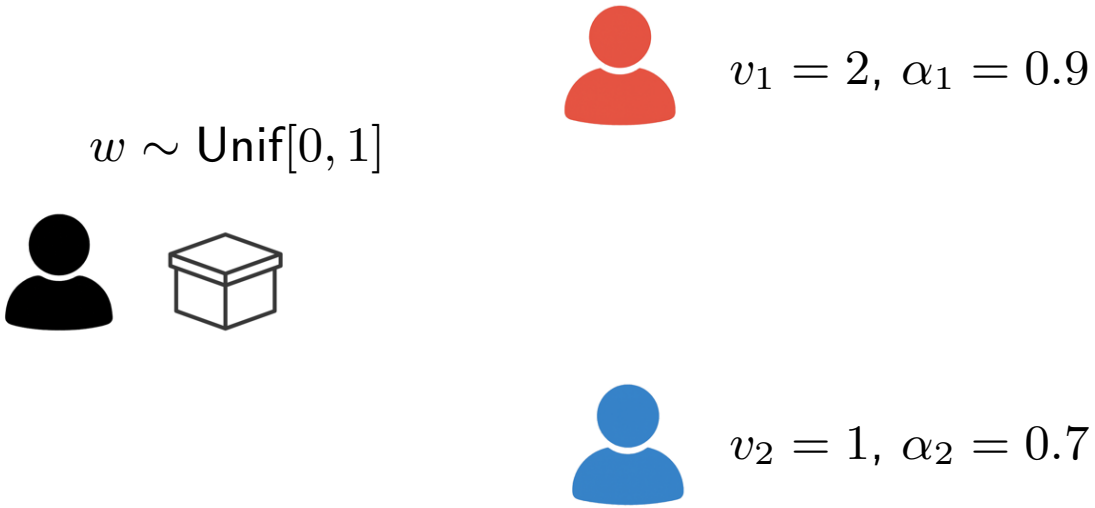

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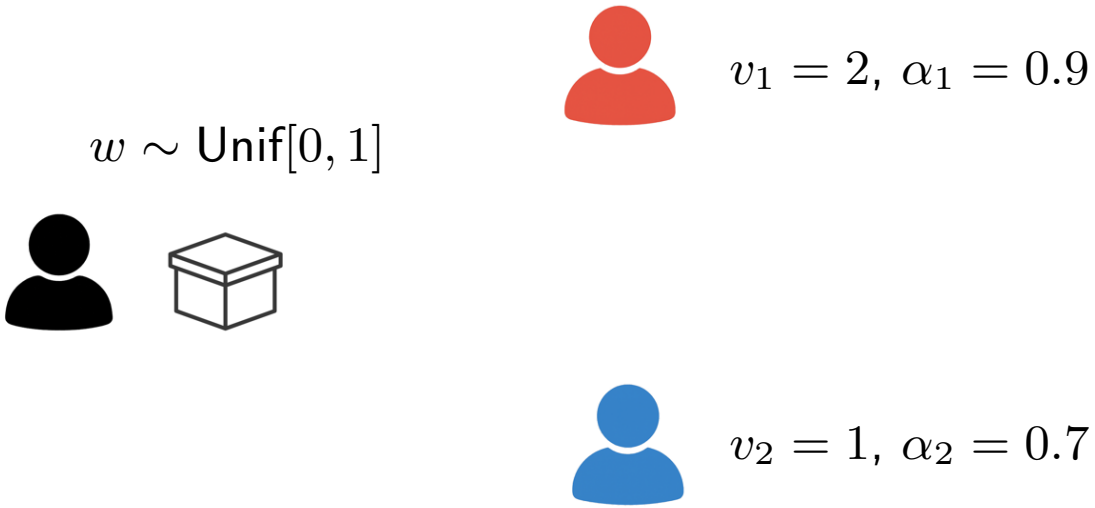

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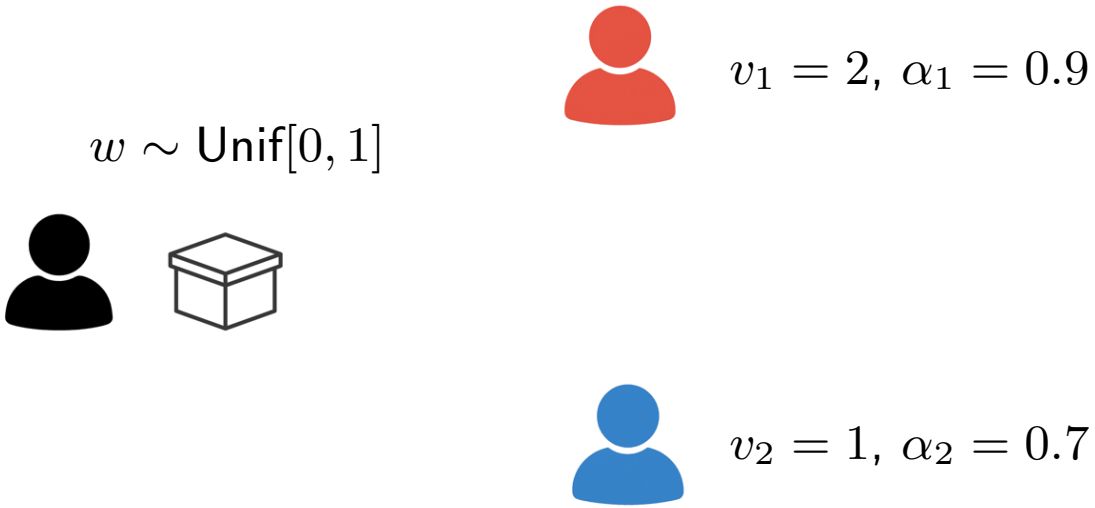

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Extensions

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- First-best relaxation:

conditional dist of w given θ

marginal dist of θ

$$\bar{V} = \max_{q(i|w,\theta)\geq 0} \quad \sum_{i=1}^{n} \int_{w\in\Omega} \int_{\theta\in\Theta} v_i(\theta)\, q(i|w,\theta)\, dG(w|\theta)\, dH(\theta)$$

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$$\sum_{i\in[n]} q(i|w,\theta) \leq 1, \qquad \forall\, w \in \Omega,\ \theta \in \Theta.$$

$q(i|w,\theta)$: prob of allocating to receiver i conditional on (w,θ)

Optimality of public persuasion under fixed ordinal ranking

Assumption 1 (fixed ordinal ranking). $v_i(\theta) \geq v_j(\theta)$ for any receivers $i < j$ and realization $\theta \in \Theta$.

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Negative result: Without Assumption 1, public persuasion may no longer be optimal (see negative example in the paper)

Numerical example

- $n = 6$ receivers. Receivers 1 and 2 are top-tier, receivers 3 and 4 are mid-tier, and receivers 5 and 6 are lower-tier.
- Linear utilities: $u_i(w) = w - \alpha_i$, with $\alpha_1 = \alpha_2 = 0.9$, $\alpha_3 = \alpha_4 = 0.8$, and $\alpha_5 = \alpha_6 = 0.7$.
- $w \perp \theta$, and w follows a discrete uniform dist on $\left\{\frac{1}{100}, \frac{2}{100}, \ldots, \frac{99}{100}, 1\right\}$.
- $v_i(\theta) \sim$ Unif$[\bar{v}_i - \epsilon, \bar{v}_i + \epsilon]$, with $\bar{v}_1 = \bar{v}_2 = 3$, $\bar{v}_3 = \bar{v}_4 = 2$, $\bar{v}_5 = \bar{v}_6 = 1$, and dispersion parameter $\epsilon \in (0, 0.5)$.

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- To facilitate numerical computation, discretize sender's preferences:
  1. Draw T independent realizations $\{\theta_t\}_{t=1}^T$ by independently sampling

     $$v_i(\theta_t) \sim \mathrm{Unif}[\bar{v}_i - \epsilon, \bar{v}_i + \epsilon]$$

     for each receiver $i \in [n]$ and each $t \in [T]$.
  2. Assume that each θ_t occurs with equal probability $1/T$: $\mathbb{P}_\theta(\theta_t) = 1/T$ for all $t \in [T]$.

Candidate public persuasion mechanism

Consider the following LP:

$$
\begin{aligned}
V^{\mathrm{P}} = \max_{q(i|w,\theta)\geq 0} \quad & \sum_{i=1}^{n} \int_{w\in\Omega} \int_{\theta\in\Theta} v_i(\theta)\, q(i|w,\theta)\, dG(w|\theta)\, dH(\theta) \\
\text{s.t.} \quad & \int_{w\in\Omega} \int_{\theta\in\Theta} u_i(w)\, q(i|w,\theta)\, dG(w|\theta)\, dH(\theta) \geq 0, \quad \forall\, i \in [n], \\
& \boxed{\int_{w\in\Omega} \int_{\theta\in\Theta} \mathbb{1}\left[v_i(\theta) > v_j(\theta)\right] u_i(w)\, q(j|w,\theta)\, dG(w|\theta)\, dH(\theta) \leq 0,} \quad \forall\, i \neq j, \\
& \sum_{i\in[n]} q(i|w,\theta) \leq 1, \quad \forall\, w \in \Omega,\, \theta \in \Theta.
\end{aligned}
$$

one additional constraint
added to first-best relaxation

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V^{\mathrm{P}} = \max_{q(i|w,\theta)\geq 0} \quad & \sum_{i=1}^{n} \int_{w\in\Omega} \int_{\theta\in\Theta} v_i(\theta)\, q(i|w,\theta)\, dG(w|\theta)\, dH(\theta) \\
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& \sum_{i\in[n]} q(i|w,\theta) \leq 1, \quad \forall\, w \in \Omega,\, \theta \in \Theta.
\end{aligned}
$$

one additional constraint
added to first-best relaxation

optimal solution of V^{P}

- **Public persuasion mechanism π^{P}:** broadcast $s = i$ with prob $q^*(i|w,\theta)$

Candidate public persuasion mechanism

Consider the following LP:

$$
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- <u>Note</u>: π^{P} is not necessarily the optimal public persuasion mechanism

Numerical results

- Set $\epsilon \in \{0.1, 0.2, 0.3, 0.4\}$ and $T \in \{20, 50, 100, 200\}$.
- For each combination of (ϵ, T), generate $N = 500$ samples for MC simulation.
- Median and maximum of the performance-loss ratio $1 - V^{\mathrm{P}}/\bar{V}$:

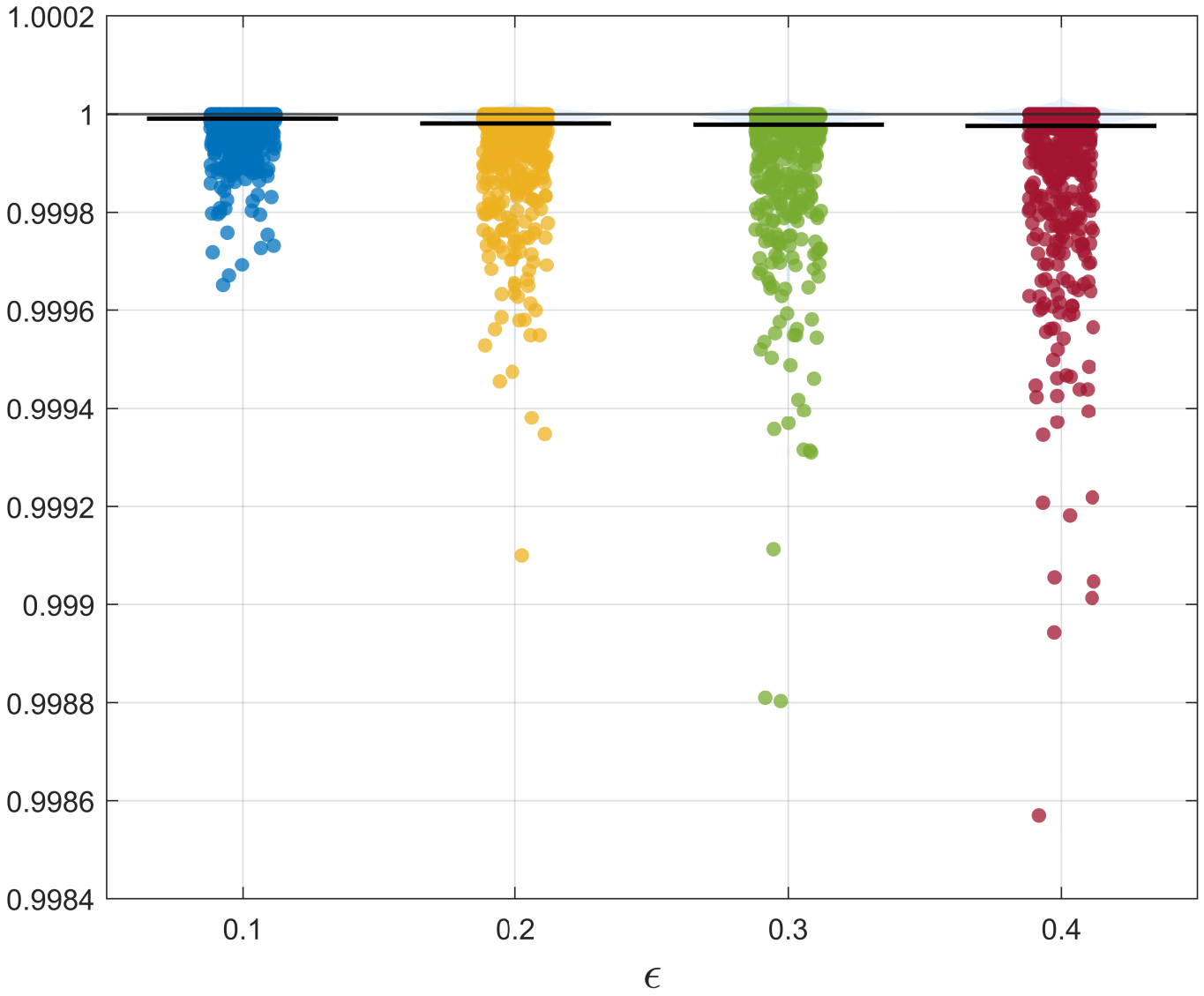
	ϵ			
T	0.1	0.2	0.3	0.4
20	1.064	2.037	2.281	2.540
50	0.581	1.132	1.415	2.034
100	0.366	0.684	0.789	1.305
200	0.206	0.404	0.485	0.758

(a) Median of $10^5 \times (1 - V^{\mathrm{P}}/\bar{V})$

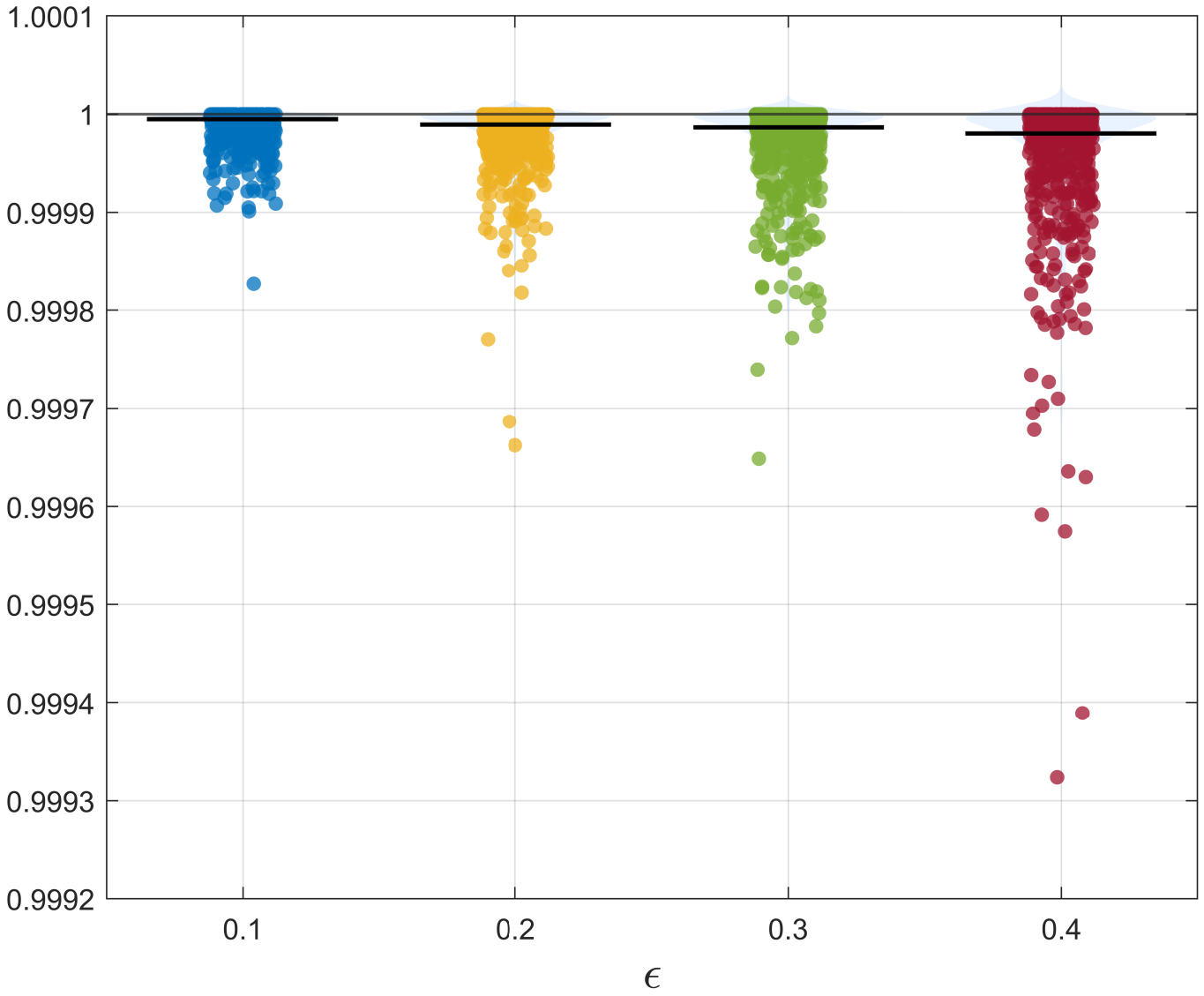
	ϵ			
T	0.1	0.2	0.3	0.4
20	3.489	8.989	11.966	14.298
50	1.732	3.379	3.517	6.762
100	0.819	1.867	1.976	2.284
200	0.277	0.625	0.611	1.302

(b) Maximum of $10^4 \times (1 - V^{\mathrm{P}}/\bar{V})$

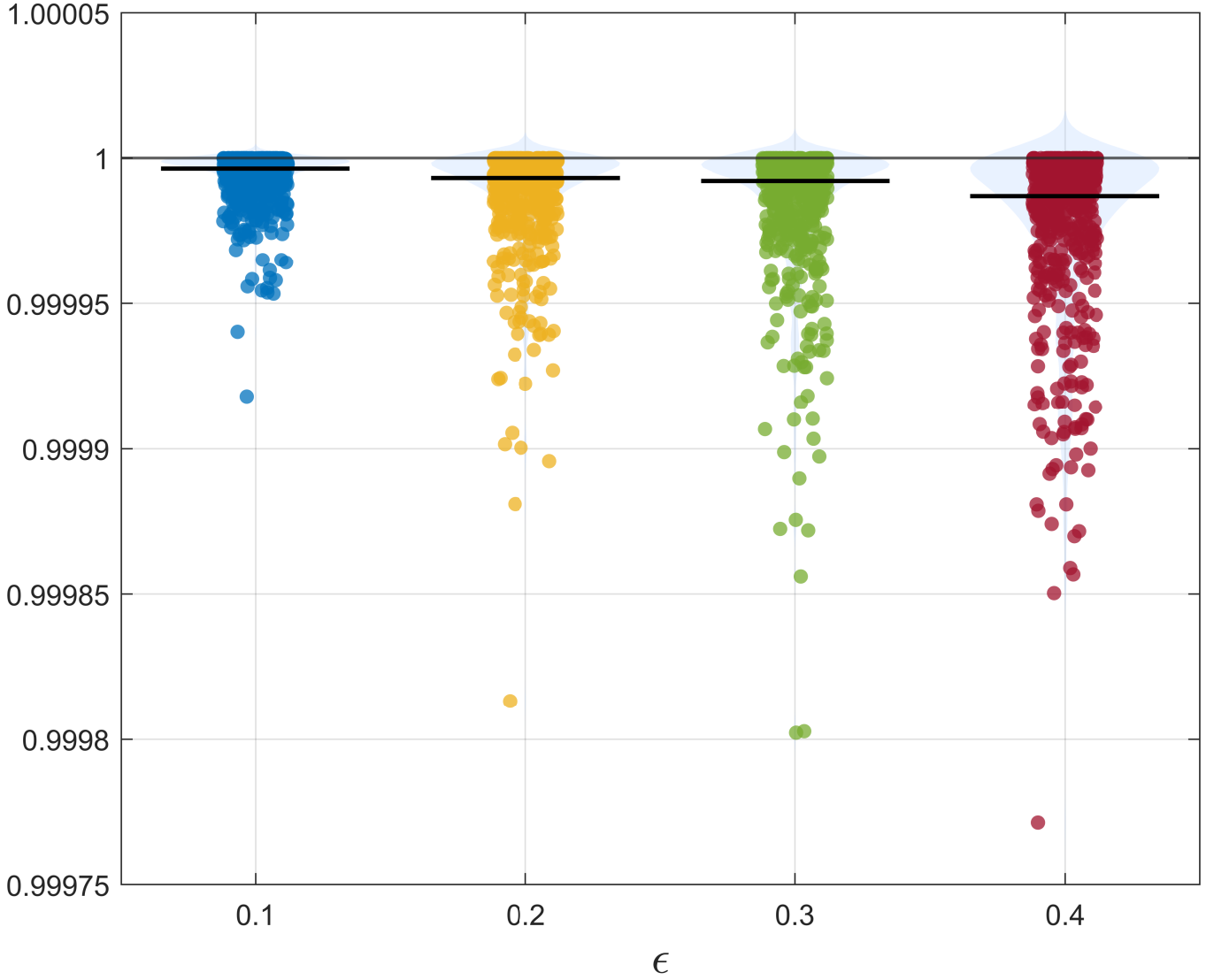
Distribution of $V^{\mathrm{P}}/\bar{V}$ $(T = 20)$



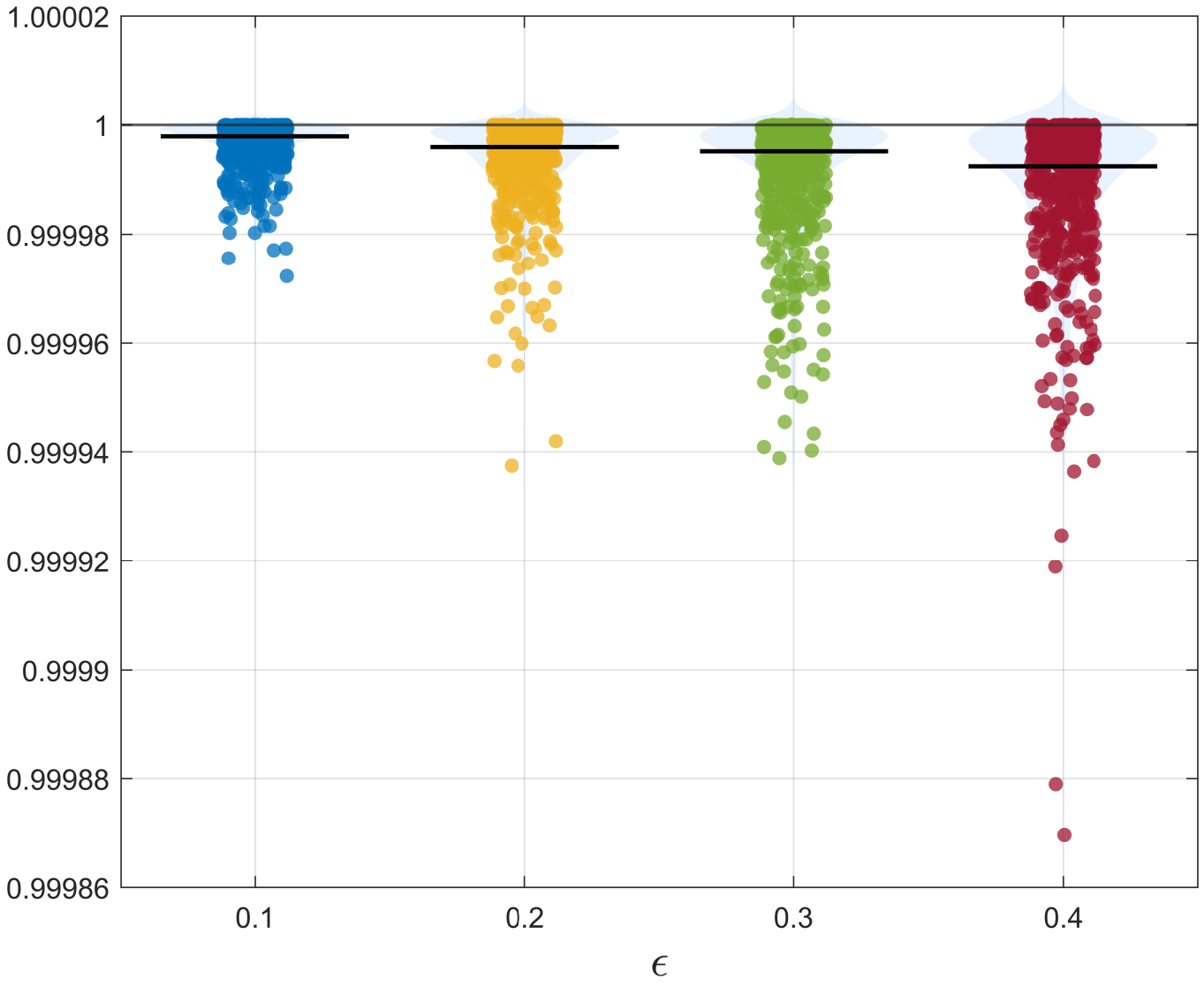
Distribution of $V^{\mathrm{P}}/\bar{V}$ $(T = 50)$



Distribution of $V^{\mathrm{P}}/\bar{V}$ $(T = 100)$



Distribution of $V^{\mathrm{P}}/\bar{V}$ $(T = 200)$



Other assumptions

- **Multiple actions:** each receiver selects from multiple actions regarding the good

  Public persuasion is optimal?

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  Counter-example: two identical goods to allocate to two receivers $\longrightarrow$ externality between receivers vanishes $\longrightarrow$ problem decouples over receivers.

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- **Key assumptions:** sender (i) allocates a **single good** and (ii) has a known **preference ordering** over receivers.

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- **Key assumptions:** sender (i) allocates a **single good** and (ii) has a known **preference ordering** over receivers.

- **Open questions:**
  (i) Other scenarios for public persuasion to be optimal
  (ii) Smoothness conditions to bound suboptimality gap/approximation ratio when public persuasion is suboptimal

Structural properties of linear-utility case

Special case: linear utilities

Utility function $u_i(w) = \kappa_i (w - \alpha_i)$ for each receiver i, where $w \in [0, 1]$

- Receivers care only about posterior mean: accept iff it exceeds α_i
- Sender equivalently optimizes dist of posterior means (we use a different approach)

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Related literature

- Extreme-point approach to characterize an optimal persuasion mechanism: [Candogan, 2022], [Kleiner et al., 2021], [Arieli et al., 2023]
- Dual approach for optimality conditions: [Dworczak and Martini, 2019]

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- **Dual approach** for optimality conditions: [Dworczak and Martini, 2019]
  - [Dworczak and Martini, 2019] study a more general problem, interpreting dual price as Walrasian equilibrium price in a persuasion economy
  - We consider a special case in which sender's utility is piecewise constant and increasing in posterior mean (as in [Candogan, 2022]), but we explicitly characterize set of optimal persuasion mechanisms by dualizing different constraints

our approach

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First-best relaxation with linear utilities:

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\max_{q(i|w)\geq 0} \quad & \sum_{i=1}^{n} v_i \cdot \int_0^1 q(i|w)\, g(w)\, dw \\
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Assumption (WLOG): receivers' hiring thresholds satisfy $0 < \alpha_n < \cdots < \alpha_2 < \alpha_1$

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First-best relaxation with linear utilities:

dualize participation constrs.
with dual variable $\mu_i \geq 0$

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Optimality conditions

Lagrangian dual problem:

$$V^{\mathrm{LR}}(\boldsymbol{\mu}) = \int_0^1 \left(\max_{\substack{q(i|w)\geq 0, \\ \sum_{i\in[n]} q(i|w)\leq 1}} \sum_{i=1}^{n} \underbrace{\Big\{ v_i + \mu_i \big(w - \alpha_i\big) \Big\}} \cdot q(i|w) \right) \cdot g(w)\, dw.$$

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allocate to the receiver with the highest positive value (if any)

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$\boldsymbol{\mu}^* \triangleq \arg\min_{\boldsymbol{\mu}\in\mathbb{R}_+^n} V^{\mathrm{LR}}(\boldsymbol{\mu})$: optimal dual variable

<u>Upper envelope function:</u> $h\big(w; \boldsymbol{\mu}^*\big) \triangleq \max_{i\in[n]} \Big\{ v_i + \mu_i^* \big(w - \alpha_i\big) \Big\} \vee 0$

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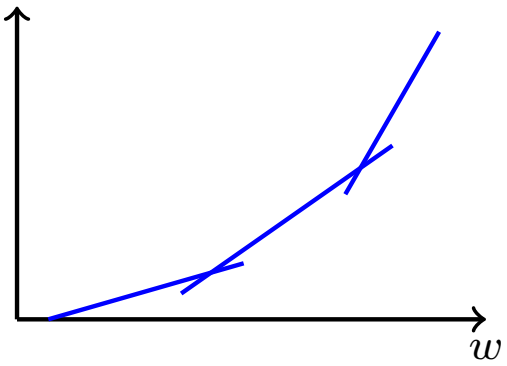
$\Longrightarrow h\big(w; \boldsymbol{\mu}^*\big)$ is convex, increasing, and piecewise linear



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Optimality conditions (informal). A (public) persuasion mechanism is optimal iff it satisfies:

1. For each linear segment of $h\big(w; \boldsymbol{\mu}^*\big)$: allocate w in this range exclusively to receivers whose point (α_i, v_i) lie on the segment
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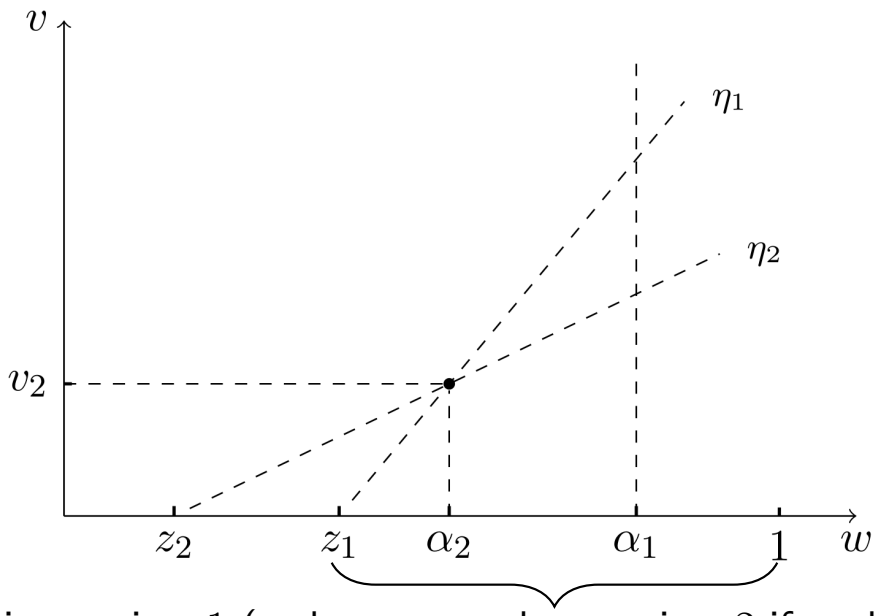
$\implies$ Problem **decouples** over segments of $h\big(w; \boldsymbol{\mu}^*\big)$

Optimal persuasion mechanisms

Two-receiver case: closed-form characterization



prioritize receiver 1 (and recommend to receiver 2 if qualified goods remain)

exclusively target receiver 2

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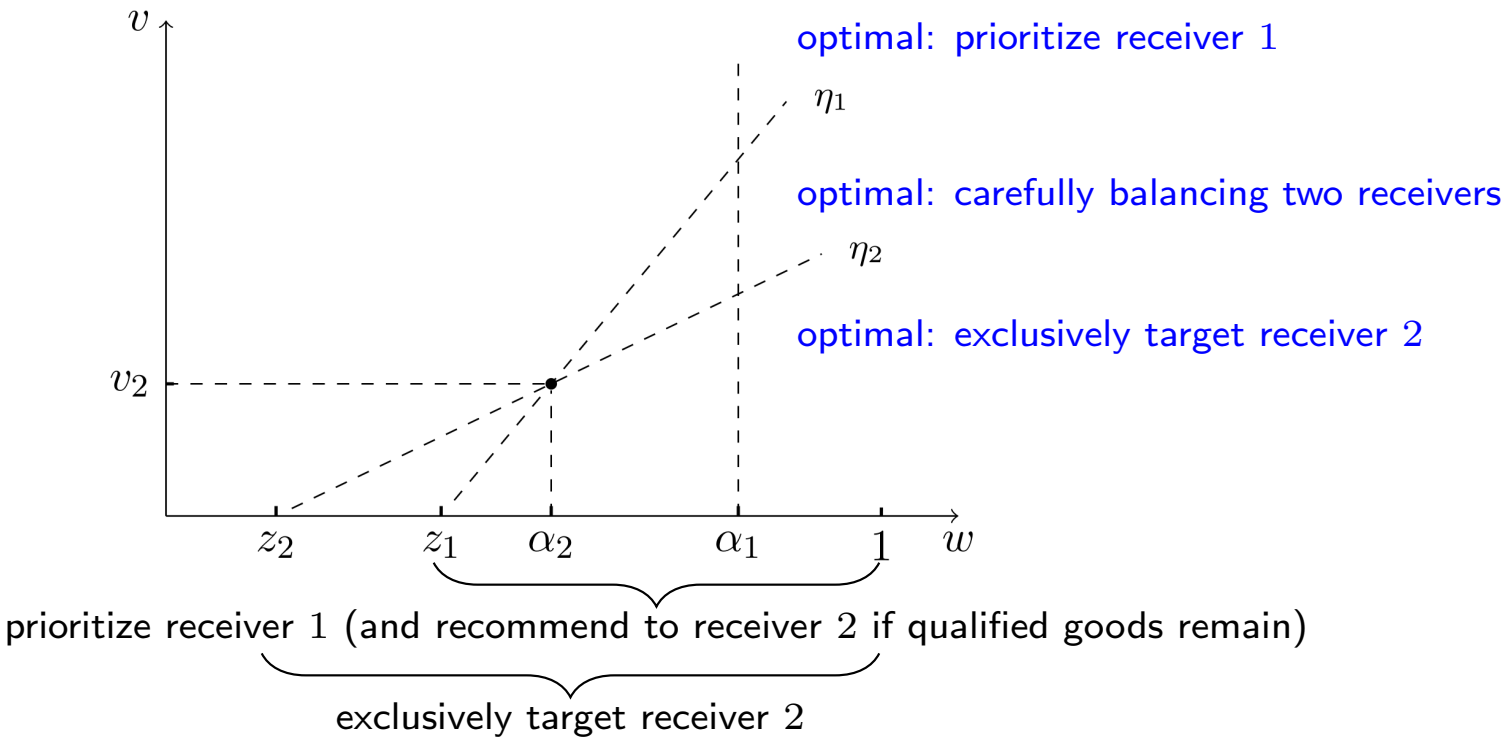

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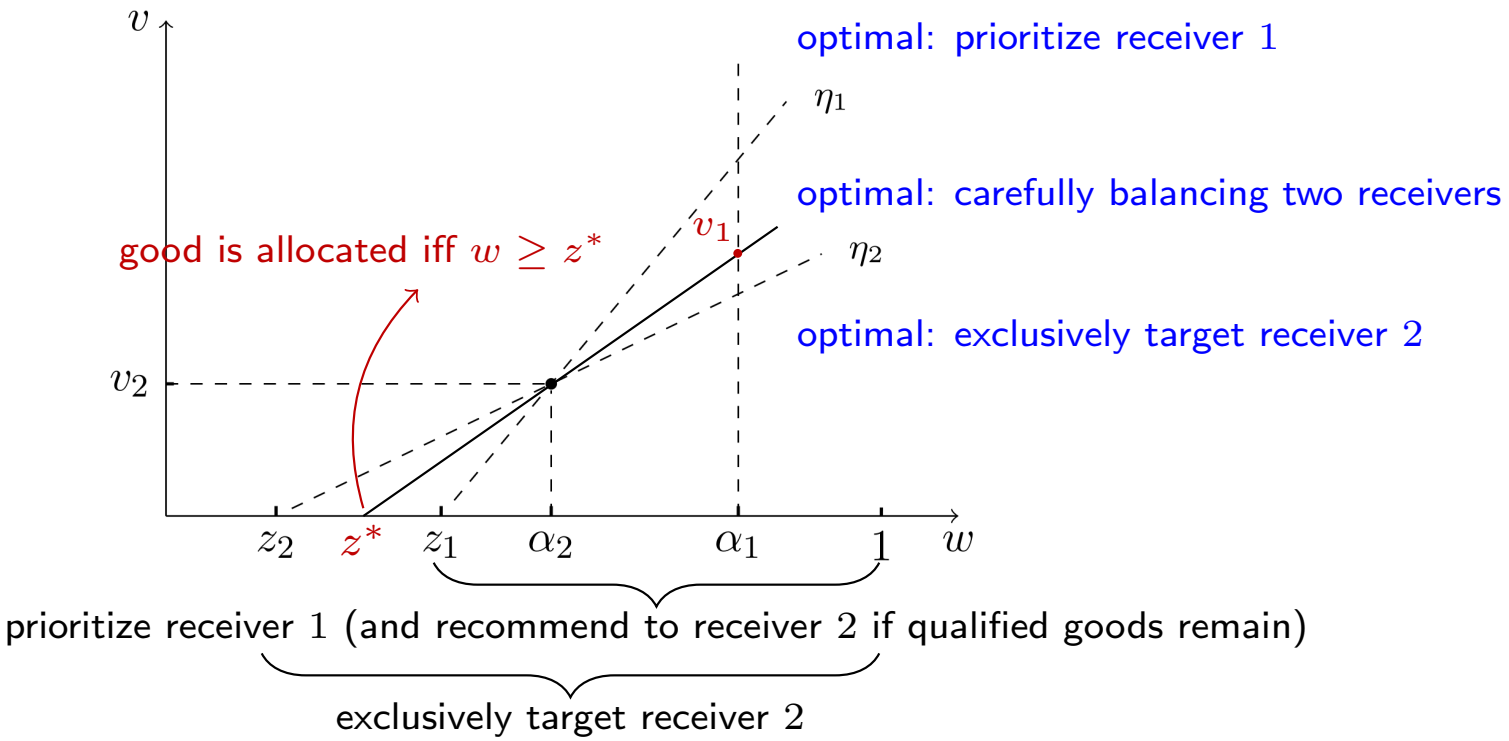

Any mechanism that satisfies the following is optimal:

1. Good is allocated if and only if $w \geq z^*$
2. Both receivers' participation constraints bind

Optimal persuasion mechanisms

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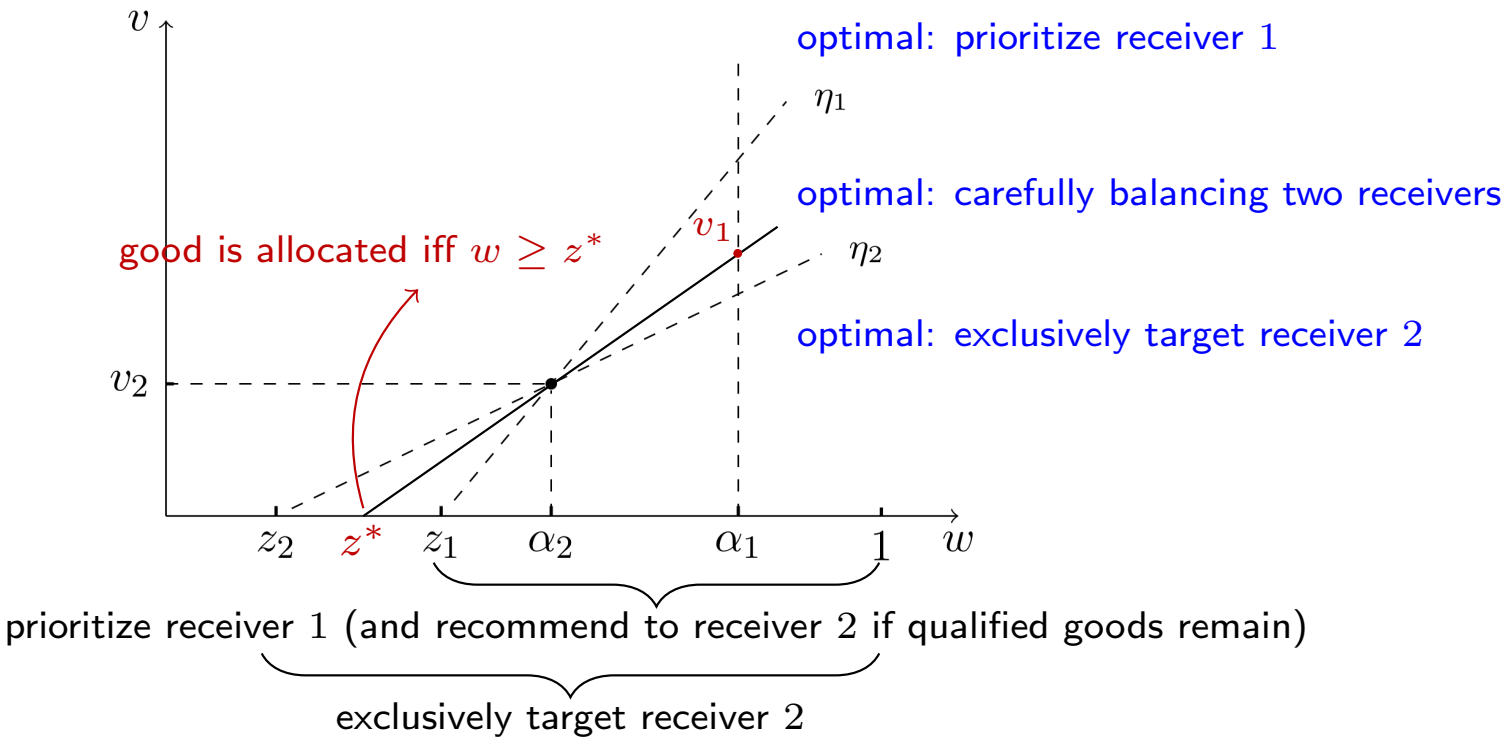

General case:

- **Complete** characterization of the upper envelope function $h(w; \boldsymbol{\mu}^*)$
- **Multiple ways** to construct optimal persuasion mechanisms

Summary

- We study single-good resource allocation in the Bayesian persuasion context, where the sender has known preferences over the receivers.
- Operational takeaway: **public persuasion** remains optimal when sender has a fixed preference order
  - Analysis is based on first-best relaxation: public persuasion obtains first-best performance.

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Reference: C. Chen and X. Qi. 2024. Optimality of Public Persuasion for Single-Good Allocation. Major Revision at *Operations Research*.

Working paper available at `https://papers.ssrn.com/abstract=4879213`

Appendix